Locator map generator

Project report

A geographic mapping automation for Cognizant Foundation India

Built 22 September to 27 October 2026

Report prepared 30 October 2026

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1. Summary

What it does	Generates geographic locator maps for CSR funding proposals
Replaces	Maps drawn by AI image tools, which invented place names
Built over	36 days, 22 September to 27 October 2026
Commits	32
Lines changed	11,719 added, 944 removed
Python written	7,940 lines across 17 files
Of which tests	1,188 lines, 134 test cases
Documentation	1,589 lines across 5 Markdown files
Administrative units covered	36 states and union territories, 785 districts, 7,146 blocks
Boundary datasets	4 used for geometry, 1 used only to cross-check
External services	3 for place search, 1 for hosting, 1 for code hosting
Runtime dependencies	9, all free and open source
Map types produced	4
Output formats per map	4, plus a combined PDF and a render log
Live deployment	https://map-cognizant.streamlit.app/
Repository	https://github.com/anuragkumar06-ship-it/Automation
Working tree on disk	1.5 GB, of which 167 MB is git history

2. The problem

Cognizant Foundation India includes a locator map in CSR funding proposals. The map has to show where a programme runs: the state inside India, the district inside the state, and the blocks inside the district, with site points for hospitals, schools or camps.

Before this tool, those maps were produced with AI image generators. One surviving example, for Madurai district in Tamil Nadu, contained eight block names that do not exist in any government register: Kallandiri, Othakadai, Ayilangudi, Keela Kallandiri, Sakkimangalam, Vellikundram, Kadakinaru and Mathur. The same map omitted four real Tamil Nadu districts (Tiruppur, Namakkal, Chengalpattu and Mayiladuthurai), misspelled Dindigul, and drew an India outline that did not match the official depiction.

A funding proposal carrying invented administrative units is a problem of accuracy before it is a problem of design. That is why the governing brief puts accuracy first, image quality second and ease of use third, and why it forbids any AI in the pipeline. The tool is deterministic: the same input produces a byte-identical output every time.

3. What the tool produces

3.1 Four kinds of map

Three-panel locator map. India with the target state marked, the state with the target district marked, and the district with its blocks and site points. The panels are joined by detail callouts in the convention an atlas uses for an inset. This is the original deliverable from the brief.

Single-state coverage map. One state with several districts marked, for showing where a programme already runs across that state. Added in October 2026 after HR supplied a Karnataka reference image showing 13 districts.

Multi-state coverage map. One continuous panel covering several states, with the districts picked in each one marked. Added 22 October 2026 after written feedback from Madhura K in Communications, who pointed out that a programme running in three states could only be mapped one state at a time, producing three separate maps.

Whole-state marking. A state with no district picked is filled to its own edge. Added 27 October 2026 at Deepak's request, for programmes that run across a whole state rather than named districts. One state on its own gives India with that state filled.

Beside a multi-state map the tool also writes one map per chosen district, showing that district's blocks drawn large and named.

3.2 Output files

Each map is written in four formats: SVG and PDF as vectors, and PNG at 300 dpi and 600 dpi. Every job also produces a single combined PDF holding all of its maps in reading order, and a `render_log.txt` recording the inputs, the validation result and the dataset versions used.

Measured output counts for the three example configurations in `maps/`:

Configuration	Files written	Why
`madurai.yaml`	18	Strip, 3 panels separately, combined PDF, log
`karnataka_coverage.yaml`	5	One map, 4 formats, log
`multi_state_coverage.yaml`	26	Overview, 5 district maps, combined PDF, log

4. Architecture

4.1 Code inventory

Measured 30 October 2026 with `wc -l`.

File	Lines	Responsibility
`locator/render.py`	2,337	All drawing, layout, projection and labelling
`locator/validate.py`	1,142	The five pre-render checks
`app.py`	933	Streamlit dashboard
`tests/test_multi_state.py`	478	Multi-state, block detail, framing, bundling
`locator/geocode.py`	452	Place-name search across three services

File	Lines	Responsibility
`locator/data.py`	405	Dataset registry and boundary loading
`locator/cli.py`	353	Command line, and the shared entry point
`locator/cache.py`	310	Builds the per-state cache from raw files
`locator/theme.py`	254	Dashboard CSS in the CF palette
`tests/test_sites.py`	253	Sites table cleaning
`tests/test_validate.py`	230	The Madurai case and the tehsil fallback
`tests/test_names.py`	227	Name normalisation and aliases
`locator/style.py`	220	Reads `style/brand.yaml`, projections
`locator/names.py`	177	Name keys and the alias table
`locator/sites.py`	153	Cleaning the sites table
`locator/__init__.py`	8	Package marker
`locator/__main__.py`	8	`python -m locator`
Total	7,940	

Application code accounts for 6,752 lines and tests for 1,188, a ratio of about 5.7 lines of code to 1 line of test.

4.2 Documentation inventory

File	Lines	Audience
`README.md`	516	Someone who has never opened a terminal
`HANDOFF.md`	492	Anyone picking the project up cold
`data/SOURCES.md`	275	Anyone auditing where a boundary came from
`DEPLOY.md`	160	Whoever hosts it next
`CLAUDE.md`	146	The original brief, the source of truth on intent
Total	1,589	

Configuration adds 140 lines of `style/brand.yaml`, 105 lines across five example map configurations, and 16 lines of `requirements.txt`.

4.3 How a render flows

A request enters through one of three front ends, and all three converge on `locator.cli.run_from_config()`. That single entry point is deliberate: the dashboard and the command line cannot drift apart, because they execute the same function.

```
front end (dashboard / CLI / Colab)
        |
        v
locator.cli.validate_config()      picks single-state or multi-state checks
        |
        v
locator.validate                   five checks, all must pass
        |
        v
locator.data + locator.cache       loads only the states the map needs
        |
        v
locator.render.render_map()        projects, lays out, labels, draws
        |
        v
SVG + PDF + PNG 300 + PNG 600 + combined PDF + render log
```

If validation fails, nothing is drawn. The `--force` flag lets warnings through but never missing data.

5. The data layer

5.1 Datasets used

Four datasets provide geometry. A fifth is kept only to cross-check block counts and is never drawn. Every one is recorded in `data/SOURCES.md` with its URL, download date, licence and attribution text.

Dataset	Provides	Rows	Licence
`SOI_States`	State and UT boundaries, Survey of India depiction	40	CC0 1.0
`LGD_Districts`	District boundaries	785	CC0 1.0
`LGD_Blocks`	Community development block boundaries	7,146	CC0 1.0
`LGD_Subdistricts`	Tehsil boundaries, fallback only	fetched on demand	CC0 1.0
`PMGSY_Blocks`	Cross-check only, never rendered	not loaded at runtime	CC0 1.0

The state layer holds 40 rows: 36 states and union territories, plus 4 rows marked disputed. The disputed rows exist so the external boundary can be drawn as India officially depicts it, including Jammu and Kashmir, Ladakh and Gilgit-Baltistan. They are excluded everywhere a user can pick a state. This was verified against the official depiction rather than assumed.

Names and codes are validated against the Local Government Directory, the government's own register. Geometry and names come from separate places on purpose: checking an LGD-derived layer against itself would confirm nothing.

5.2 The cache, and why it exists

Raw downloads total 438 MB across 6 files and are not committed. Running `python -m locator prepare` turns them into `data/processed`, which is 104 MB and is committed, so the application deploys without a 1.5 GB preparation step on the host.

Cache component	Files	Size
`blocks/`	36	72 MB
`districts/`	36	27 MB
`subdistricts/`	1	3.8 MB
`states.parquet`	1	874 KB
`units_index.parquet`	1	79 KB
`index.parquet`	1	19 KB

The working tree on disk is 1.5 GB in total, which includes the uncommitted raw downloads and rendered output. The git history itself is 167 MB, most of it the committed cache.

The two index files carry names and codes with no geometry at all, which is what lets the dashboard populate its dropdowns without loading a single shape. That mattered for a measured reason: loading the national district and block layers at startup used 1,016 MB, which would have exceeded what the hosting plan allows and killed the application on its first boot. After the split it uses 110 MB.

Compression is zstd at level 15. State geometry is simplified to 150 m; district and block geometry passes through unchanged.

The prepare step also repairs self-intersecting outlines using Shapely's `make_valid()`. It repaired 297 polygons across all layers. The largest area change measured was 0.000000 per cent, at most 0.2 square metres: the crossing is untangled and no boundary moves.

5.3 Supporting tables

Table	Rows	Purpose
`data/aliases.csv`	51	Spelling variants, e.g. Thirumangalam to TIRUMANGALAM
`data/lgd/reference_counts.csv`	4	Independent counts for strict validation
`data/sea_labels.csv`	12	Names of seas and gulfs for the state panel

5.4 Known gaps in the data

Seventeen districts have no published block boundaries. For fourteen of them, all in Rajasthan, tehsils were recovered by spatial containment at 100 per cent coverage. Those are labelled tehsils everywhere they appear, never blocks, because the brief is explicit that a taluk must never be called a block.

`reference_counts.csv` holds only four rows: districts in Tamil Nadu, districts in Bihar, blocks in Madurai and blocks in Bhagalpur. Checks 2 and 3 are therefore strict only in those four places. Elsewhere they report that no independent reference is recorded rather than silently passing, which is a deliberate choice: the map says what has been verified and what has not.

6. Validation

Five checks run before anything is drawn. They live in `locator/validate.py`.

1. State, district and block names exist in the boundary layer, resolved through the alias table.
2. The district count in the state layer matches the LGD reference, with any missing or extra name reported by name.
3. The block count in the district layer matches the LGD reference, reported the same way.
4. Every site point falls inside the stated district, and the block it actually falls in is reported. A mismatch with what the user said is a warning, never a silent correction.
5. Geometries are valid, with no self-intersections and no gaps beyond tolerance.

Three structural decisions sit underneath these checks, and each exists because a name-based approach failed.

Districts join to states by LGD code, not by name. The state layer writes "DADRA & NAGAR HAVELI & DAMAN & DIU" where the district layer writes "DADRA,NAGAR HAVELI,DAMAN & DIU", and a name join silently loses that union territory. All 785 districts match a state by code.

A multi-state map marks districts by LGD code for the same class of reason. A district name is only unique within its own state: Bilaspur is in both Himachal Pradesh and Chhattisgarh, and Hyderabad is a district of Telangana but not of Karnataka. A name-based mask would mark the wrong district while all five checks still passed. There is a test for this.

Blocks to mark are keyed by district in the configuration, because a block name is only unique inside its own district.

6.1 The lesson that shaped the project

The coordinate for Government Rajaji Hospital in `maps/madurai.yaml` was originally typed from memory. It sat 1,482 metres from the real hospital. All five validation checks passed, because the point was still inside Madurai district and still inside a plausible block. Nothing caught it.

That is the clearest argument for the no-AI rule in the brief. A coordinate that validates is not a coordinate that is correct. Site coordinates now come from a Google Maps right-click or from a geocoding service that returns a real record, never from recall.

7. Rendering

7.1 Projections

The India panel uses Lambert Conformal Conic fitted to India. State and district panels use the local UTM zone chosen by `estimate_utm_crs()`. Raw latitude and longitude is never plotted.

A multi-state panel uses the conic rather than UTM. Three states can span a third of the country, and a single UTM zone visibly stretches whatever falls near its edges.

7.2 Framing

States scattered down the country produce a tall, narrow bounding box. Framed on that alone, the map printed as a ribbon down a third of the sheet with the surrounding land cut to a rectangle. `_widen_extent()` opens the frame sideways to the page's own proportions and never past the country's bounds. Once the frame has opened past 0.8 of the country's width, it shows the country whole rather than slicing the outline across the north, which matters for anything published with an India map on it.

7.3 Labelling

The three-panel and single-state maps use adjustText. Its default one-second time limit makes output depend on how busy the machine is, so `iter_lim` is pinned at 260 iterations to keep renders reproducible.

The multi-state map uses a purpose-built placer, `_label_callout_map`. Each label is tried on rings around the thing it names, nearest first, and may not touch a chosen state's outline, a covered area, a dot, a site marker, the scale bar, the north arrow or another label. Crossing another border or the coast is allowed but carries a cost, so a clear position wins when one exists. It is deterministic by construction. Started on the shape, as adjustText does, labels sat across coastlines and state outlines.

A covered district too small to see at national scale, such as Kolkata, also gets a dot so it cannot vanish.

7.4 Scale thresholds

Three thresholds decide what is drawn at what size, all measured in points on the printed page.

Constant	Value	Effect
SUBLAYER_MIN_PT	6.0	Block boundaries appear on the overview only above this
SMALL_MARK_PT	6.0	Below this a covered district also gets a dot
RIM_MIN_PT	3.0	Outline parts smaller than this are skipped

The block threshold exists because of a measurement. The median block across a sample of ten districts is 15.9 km across. That is 1.3 mm on a multi-state sheet, 5.9 mm on a single-state panel, and 40 mm on a map of its own district. A block name needs roughly 12 to 15 mm. Blocks are therefore only legible at district scale, which is why each district gets its own map, and why block boundaries are drawn unnamed on the overview and dropped entirely at national scale, where the white lines buried the district's own fill.

7.5 Brand

Every colour, font, marker and size lives in `style/brand.yaml`: 18 colours, 14 site types each with a distinct marker, 9 type sizes and 8 line widths.

Text is midnight blue `#000048`. Non-target units are light grey `#D0D0CE` with white hairline borders. The standard highlight is teal `#05819B`. Water is white. Sentence case everywhere, left-aligned titles, no gradients, no shadows and no dashed lines. Colours come from the CF brand visual identity guidelines of January 2025 and the communication guidelines of September 2024.

The multi-state map uses three strengths of the one teal, after HR reported that grey states next to grey country looked awkward. A chosen state is washed in `#D2E8ED`, its covered districts sit on top in `#034F62`, and covered blocks stay `#06C7CC`. The deep shade was added because the standard highlight sat too close in tone to the pale wash and the districts disappeared into their own state. This applies only to the multi-state map; the other map types have no pale wash to compete with and keep the standard highlight.

8. The three front ends

Streamlit dashboard (`app.py`, 933 lines). How everyone actually uses it. States and districts come from dropdowns populated out of the geometry-free index, so nothing has to be typed or spelled. Sites can be entered as coordinates or found by name. Each session gets its own temporary output directory, so two people rendering at once do not overwrite each other.

Command line (`python -m locator`). Four subcommands: `check` runs validation and draws nothing, `render` validates then draws, `fetch` downloads the boundary data ahead of time, and `prepare` builds the cache from raw files.

Colab notebook. A form interface for users who would rather not install anything. Built and working, though less polished than the dashboard.

9. External services and platforms

Eight external services are involved across development, hosting and runtime.

Service	Used for	Cost	Required
GitHub	Code hosting, data releases	Free	Yes
Streamlit Community Cloud	Hosting the live dashboard	Free	For the live site
Google Colab	Notebook front end	Free	No
Nominatim (OpenStreetMap)	Place-name search, first choice	Free	No
Photon (Komoot)	Place-name search, fallback	Free	No
Google Places API	Place-name search, last resort	Paid	No, off by default
Google Maps	Where users copy site coordinates	Free	No
Local Government Directory	Name and count reference	Free	Reference only

Place search runs as a cascade in `search_best()`: Nominatim, then Photon, then Google Places, stopping at the first usable answer. Nominatim's usage policy requires an identifying User-Agent and a request interval, so the client sends one and waits 1.1 seconds between calls. Google Places stays off unless a `GOOGLE_MAPS_API_KEY` is supplied, which keeps the free-and-open-source constraint intact.

Every result is filtered by the caller's predicate, because Photon's bounding box is a hint rather than a filter. Nominatim is called with a viewbox bounded to the chosen district, which is what stopped a search for a place in Jammu returning a result in Srinagar.

10. Dependencies

Nine runtime dependencies, all free and open source. No paid API key is needed to run the tool.

Package	Declared	Installed	Used for
geopandas	>=1.0,<2.0	1.1.4	Vector data handling
shapely	>=2.0,<3.0	2.1.2	Geometry operations
pyproj	>=3.6	3.8.0	Projections
pyarrow	>=15.0	25.0.1	Reads GeoParquet, including zstd

Package	Declared	Installed	Used for
matplotlib	>=3.8	3.11.2	Rendering and PDF output
adjustText	>=1.1	1.4.0	Label placement
PyYAML	>=6.0	6.x	Config and brand files
pandas	>=2.0	3.0.6	Tabular data
streamlit	>=1.40	1.64.0	The dashboard

Running on Python 3.12.10. Place-name search uses `urllib` from the standard library, so it adds no dependency. The combined PDF uses matplotlib's own `PdfPages`, chosen over a PDF library for the same reason.

11. Testing

134 test cases, written as 118 test functions across 4 files. The difference is parametrised tests, where one function runs against several inputs.

File	Cases	Functions	Covers
`test_multi_state.py`	42	42	Multi-state validation, block detail, framing, bundling
`test_sites.py`	36	25	Sites table cleaning and coordinate handling
`test_validate.py`	29	24	The Madurai acceptance case, tehsil fallback
`test_names.py`	27	27	Name keys, aliases, normalisation

The suite runs in about 28 seconds against the real boundary data, not against fixtures. If the boundary data is absent the module skips rather than failing, so a fresh clone can still run the rest.

Several tests encode specific past failures. One asserts that each of the eight block names invented by the old AI map is rejected. Another checks that Bilaspur in Himachal Pradesh and Bilaspur in Chhattisgarh resolve to two different LGD codes. A third checks that a block of one district cannot be marked on another.

12. Measured performance

Render times measured on the development machine, 30 October 2026, as a full cold `python -m locator render` including interpreter startup and data loading.

Map	Time	Files
Three-panel locator (Madurai)	19.5 s	18
Single-state coverage (Karnataka)	6.1 s	5
Multi-state coverage, 5 districts	17.4 s	26

Two optimisations account for most of the difference from where it started. Computing exact distances against full-resolution coastlines took 63 seconds of an 80 second render; simplifying neighbours to 500 m and using a spatial index brought the whole render to 14 seconds. Splitting the national files into a per-state cache cut startup memory from 1,016 MB to 110 MB.

13. Development history

32 commits between 22 September and 27 October 2026, across six working days.

Date	Commits	Theme
22 Sep 2026	8	Scaffold, first working map, reproducibility, brand
25 Sep 2026	5	GRH coordinate fix, per-state cache, place search, deployment
28 Sep 2026	5	Sites table fixes, 14 site types, geometry repair, memory
29 Sep 2026	5	Coverage maps, panel-per-file output, titles, deploy hardening
22 Oct 2026	6	Handoff doc, Colab fix, multi-state maps, block maps, combined PDF, colour
27 Oct 2026	3	National-scale labelling, whole-state marking, documentation

All 32 commits carry a `Co-Authored-By` trailer recording the model used: 29 for Claude Opus 5 and 3 for Claude Opus 5.5. The work was done in a conversational development loop, with the brief in `CLAUDE.md` acting as the standing specification and rendered output reviewed after each visual change.

13.1 How requirements arrived

The project did not start with a complete specification. Four features came from named stakeholders after the tool was already in use:

Request	From	Date	Result
Coverage map for a whole state	HR, with a Karnataka reference image	Oct 2026	Single-state coverage map
Maps spanning several states	Madhura K, Communications	22 Oct 2026	Multi-state coverage map
Blocks visible on those maps	Madhura K, Communications	22 Oct 2026	Per-district block maps
Mark whole states with no district	Deepak	27 Oct 2026	Whole-state marking

Two further changes came from direct observation of the tool in use: the states on a multi-state map reading as grey and looking awkward, and downloads clearing the rendered maps off the page.

14. Defects found and fixed

Fifteen distinct defects were diagnosed and fixed during development. Each is recorded with its cause, because the cause is what stops it returning.

Symptom	Cause	Fix
Same input gave different output	adjustText's 1 s time limit	Pinned `iter_lim`
80 second renders	Exact distance on full-resolution coastlines	Simplify plus spatial index
Dadra & Nagar Haveli missing	Districts joined to states by name	Join by LGD code
Sites showing "Nan"	`float('nan')` is truthy, so `str(v or "")` gave `"nan"`	Explicit NaN handling
Sites rejected as outside Kota	Seeded default plus a data editor state loop	Schema reset and delta tracking
Jammu search returning Srinagar	Unbounded Nominatim query	`viewbox` plus `bounded=1`
Geometry warning on Kaduthuruthy	185 invalid block polygons	`make_valid()` at cache build
Application killed on deploy	1,016 MB at startup	Geometry-free index files
"Easting [metre]" printed on panels	`set_axis_off()` dropped when frames were added	Explicit `set_xlabel("")`
Coverage map 10 inches tall	Fitted one dimension only	Fit both, then trim the page
KeyError on the live site	New `app.py` against a cached old `cli.py`	Defensive reads plus reboot
Maps vanished after one download	Results drawn inside `if go:`	Result kept in `session_state`
Covered districts a white scribble	Block boundaries drawn at any size	`SUBLAYER_MIN_PT` threshold
Dark specks over Tamil Nadu and the Bengal coast	State rim drawn round enclaves and islands	Outer-edge-only outlines
Labels across coastlines and outlines	adjustText starts each label on its shape	Purpose-built callout placer

Three cells of the Colab notebook also contained a raw newline where an escaped one was intended, which split Python string literals across lines and made those cells fail to compile. Found by compiling every cell rather than by running the notebook.

15. Constraints the project worked under

These come from the brief and were treated as non-negotiable.

No AI anywhere in the pipeline: no language model call, no image generation, fully deterministic output. The tool was built conversationally with an AI assistant, but nothing in the shipped pipeline calls one.

Never draw, approximate or invent a boundary. Every polygon comes from a sourced dataset. If data is missing the tool stops and says so rather than substituting a guess.

Never label a taluk as a block. Where block data does not exist, the fallback is clearly labelled as tehsils on the face of the map.

Free and open source only, with no paid API key required to run.

No account creation or password entry. Where a portal requires registration, the instructions tell the user to download the file manually and say where to put it.

Users are non-technical and work in Google Workspace, so the README is written for someone who has never opened a terminal.

16. Open items

1. Reboot the Streamlit app. The live site needs a manual restart from share.streamlit.io before the most recent work is visible, because an auto-redeploy reruns the script without reliably reloading modules already imported.
2. `reference_counts.csv` covers four entries. Filling in more states would make checks 2 and 3 strict in more places.
3. CF approval for logo use, now that the repository is public. The communication guidelines require approval per use.
4. From the brief, not started: a CF programme layer read from a CSV exported from a Google Sheet, to plot existing CF work and list programmes in the same or adjacent districts.
5. Streamlit Community Cloud sleeps an idle application. `DEPLOY.md` records the route to an internal Cognizant server if the tool is adopted permanently.

17. Deliverables produced

Deliverable	Form	Audience
The tool itself	Python package plus Streamlit app	CF proposal writers
`README.md`	516 lines	Non-technical users
`HANDOFF.md`	492 lines	Next developer
`data/SOURCES.md`	275 lines	Anyone auditing a boundary
`DEPLOY.md`	160 lines	Whoever hosts it
`Locator map generator - how it works.docx`	1,868 words, 2 figures, 5 tables	HR, for company-wide presentation
Example configurations	5 YAML files	Users, as starting points
This report		Internship submission

18. Acceptance against the brief

Case 1, Madurai. Tamil Nadu, Madurai district, Madurai East block, with Government Rajaji Hospital as a site. Madurai shows 13 blocks, matching the LGD list. None of the eight invented block names appear. All four previously missing Tamil Nadu districts appear, and Dindigul is spelled correctly. The hospital falls in Madurai West, not Madurai East as originally stated, and the tool reports that rather than forcing the user's answer.

Case 2, Bhagalpur, Bihar. Configuration at `maps/bhagalpur.yaml`. Bhagalpur shows 16 blocks, matching the district administration's published list.

Both cases pass all five checks with no errors and no warnings.